QUANTUM ARENA 1.0

TITLE PAGE

- **Problem Statement Title -** "Excessive electricity is wasted on campus due to unnecessary use of lights and appliances. A smart IoT-based system is needed to reduce power wastage and improve energy efficiency."
- **Theme -** Smart Energy Optimization and Sustainable Campus Management using IoT.
- **PS Category -** Hardware + Software.
- **Team Name –** Democracy Defenders.

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IDEA TITLE

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Our solutions aims to build a sustainable future by reducing energy waste and promoting intelligent

1. **Smart Energy Monitoring** – Real-time tracking of electricity usage.

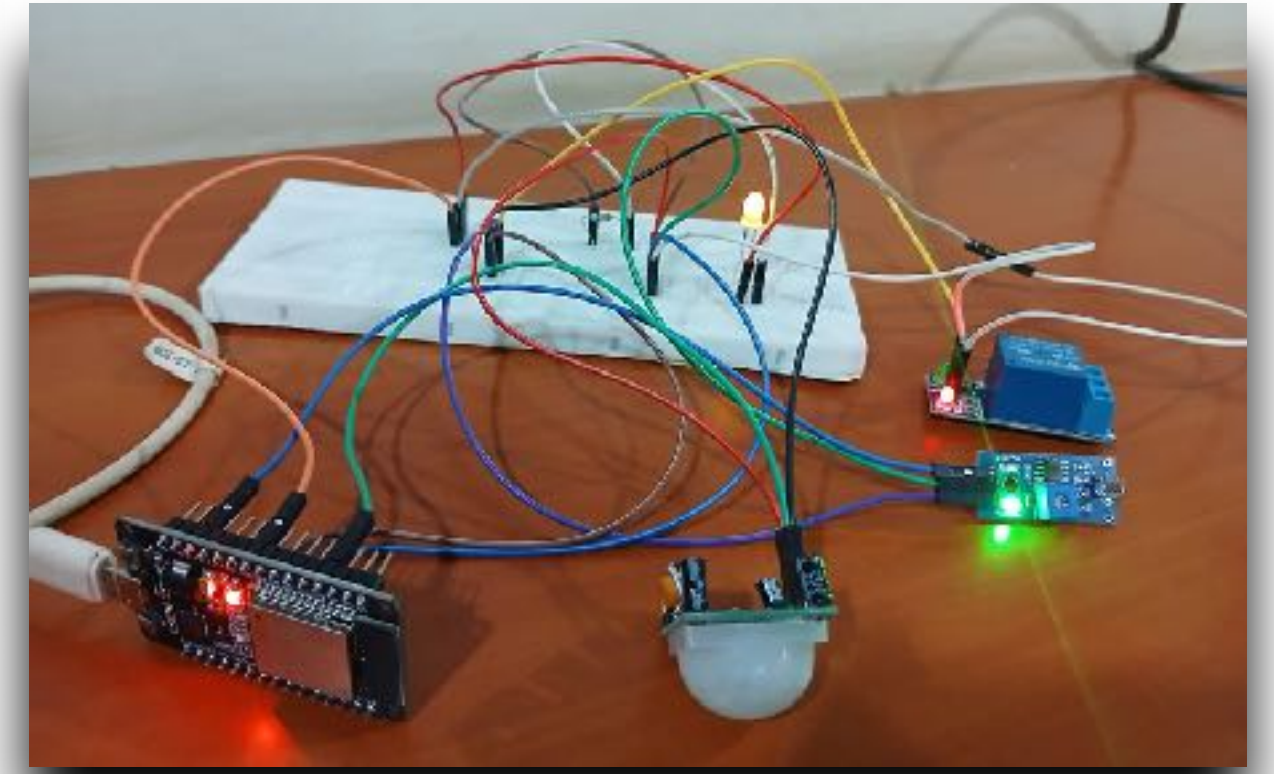
2. **AI-Based Optimizations** – Predicts and reduces unnecessary energy consumption.

3. **IoT Integration** – Sensors (PIR, LDR) automate device control.

4. **Cost Reduction** – Minimizes electricity bills using smart scheduling.

5. **Energy Efficiency** – Reduces power wastage in Campus.

6. **Automated Control System** – Lights & appliances controlled based on environment



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TECHNICAL APPROACH

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METHODOLOGY

LDR Sensor
Detects light intensity (light/dark condition).

ESP32 Processing
Reads sensor data and makes decision.

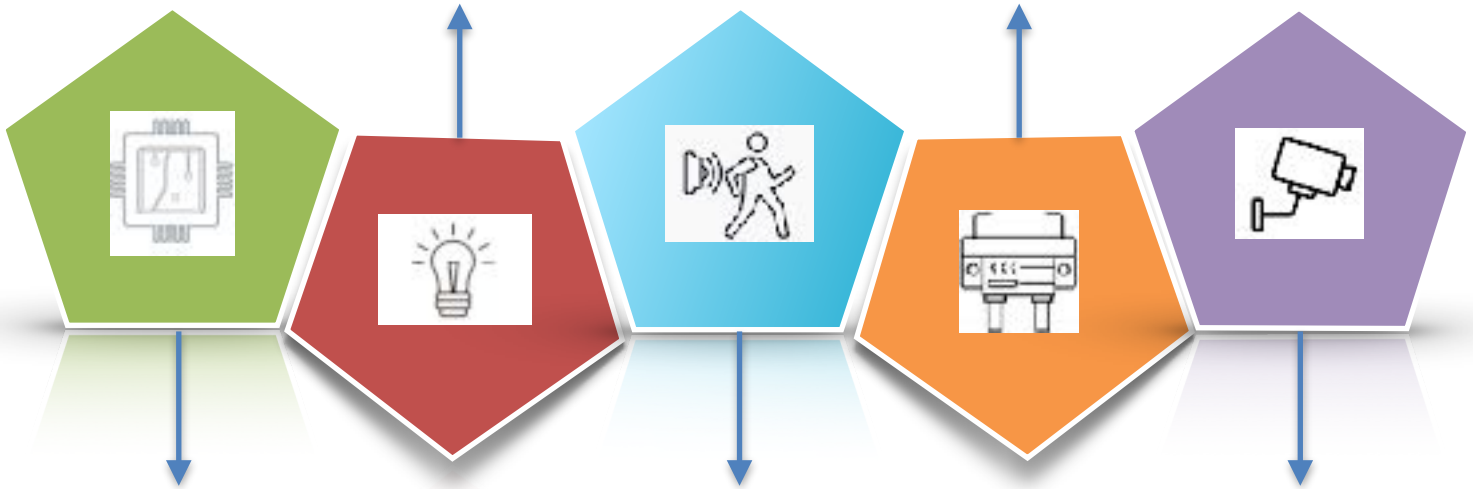
PIR Sensor
Detects human motion (presence of person).

Relay Module
Controls light ON/OFF based on signals.

Output / Smart Control
Light turns ON/OFF automatically and saves energy.

Light Dependent Resistor
Detects light intensity to automate light control.

Relay Module
Relay isolates low voltage(ESP32) and controls high voltage load.



Microcontroller Processing
ESP32 reads signal and applies control logic.

PIR Sensor
PIR sensor detects changes in human body heat (IR radiation).

CCTV Camera Sensor
CCTV camera provides real-time monitoring & improving security.

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FEASIBILITY AND VIABILITY

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FEASIBILITY

TECHNICAL FEASIBILITY

1.ESP32-based IoT system enables real-time control
2.Integration of sensors (motion, energy monitoring) is scalable
3.Web dashboard ensures easy remote access
4.Low latency communication using WiFi

ECONOMICAL FEASIBILITY

1.Low-cost components (ESP32, relay, sensors)
2.No expensive infrastructure required
3.Suitable for large-scale deployment camps

TIME MANAGE FEASIBILITY

1.Simple ML models like Linear Regression require less training time
2..Use of ready libraries (Pandas, Scikit-learn, Streamlit) speeds up development
3..Modular approach (data → model → output) allows quick implementation

OPERATIONAl FEASIBILITY

1.Easy to use dashboard (user-friendly UI)
2.Automatic + manual control system
3.Minimal maintenance required
4.Works with existing electrical systems

VIABILITY

1.Real-World Impact
Reduces energy wastage significantly
Supports smart home & smart building systems
Promotes sustainable energy usage

2.Scalability
Can be expanded to multiple rooms/buildings
Cloud integration possible (future scope)
Can connect with mobile apps

3.Reliability & Security
Controlled access via network
Can integrate authentication (future)
Stable performance with ESP32

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IMPACT AND BENEFITS

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Impact

1.Reduced Energy Consumption
Minimize unnecessary energy usage and wastage.

2. Peak Load Reduction
Helps in lowering electricity demand during peak hours.

3.Environmental Sustainability
Reduces carbon emissions and supports eco-friendly living.

4.Cost Efficiency
Decreases electricity bills for users.

5.Improved Operational Efficiency
Enhances performance in campus.

6.Smart Decision Making
Uses AI and data for better energy management.

7. Support for Smart Cities
Contributes to development of smart grids and smart infrastructure

Benefits

- Energy Saving
- Cost Reduction
- Automation
- Real-Time Monitoring
- Scalability
- Accuracy

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RESEARCH AND REFERENCES

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Research

https://ieeexplore.ieee.org/document/10167883

https :://share.google/Nzj0HHcAx0oujCSi5

https://www.ijcrt.org/papers/IJCRT24A5396.pdf

https://www.ijirset.com/upload/2025/december/84_Energy%20Consumption%20Prediction%20for%20Smart%20Campus.pdf

References

IEEE Research Paper on Smart Energy Monitoring

IJCRT Journal Paper on Energy Optimizations

ESP32 Official Documentation

Arduino Official Website

IoT Sensor Tutorials (LDR, PIR)

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REQUIREMENTS & FUTURE SCOPES

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Hardware Requirements
ESP32 / Arduino (controller)
Sensors (Motion sensor, Light sensor)
Relay Module (devices control sathi)
Jumper Wires
Breadboard
Power Supply

Software Requirements
Python Programming
Arduino IDE / Thonny
IoT Platform (Blynk / ThingSpeak)
Mobile App / Web Dashboard

Network Requirements
Wi-Fi Connection
Internet access for real-time monitoring

User Requirements
User should know basic system operation
Access to mobile or PC for monitoring

System Requirements
Laptop / Computer
Basic programming knowledge
Installed drivers and libraries

FUTURE SCOPE

- AI-Based automation
- Mobile app & Remote controller
- Voice assistance integration
- Advanced security system
- Cloud storage & analytics
- Scalability

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THANK YOU